GraphVerify: Reducing Hallucination through Knowledge Graph Fact Verification

Lena Okafor, Ravi Menon

Northbridge University

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Abstract

GraphVerify performs post-hoc Fact Verification for answers produced by a Large Language Model. It splits an answer into atomic claims, retrieves supporting evidence with SubGround from a Knowledge Graph, and uses a Relational Fact Lattice to locate matching facts quickly. Unsupported claims are rewritten or flagged rather than silently retained. On FactProbe, GraphVerify reduces the hallucination rate from about 18% to 7%; on HotpotQA it preserves Exact Match while improving claim-level reliability. The design combines retrieval, structured evidence, and a final verification pass, making it useful when a fluent answer must be audited after generation.

Keywords

Fact verification, Hallucination Reduction, Subgraph grounding, HotpotQA

1. Introduction

Hallucination Reduction requires more than retrieving context: a generated answer may contain one unsupported claim among several correct ones. FactProbe [1] provides a direct measurement. SubGround [2] supplies query-specific evidence subgraphs, while the Relational Fact Lattice in [3] offers an overlooked indexing idea for fast structured lookup. GraphVerify connects these components through post-hoc Fact Verification.

2. Method

The verifier first segments an answer into atomic claims using punctuation, conjunctions, and a lightweight dependency heuristic. Each claim is converted to a relation-oriented lookup key. SubGround retrieves a compact Knowledge Graph neighborhood, and the Relational Fact Lattice indexes candidate subject-relation-object matches for constant-time bucket access. A claim scorer combines lexical alignment, entity agreement, relation confidence, and source recency. Claims above a support threshold are retained; borderline claims are marked with uncertainty; unsupported claims are rewritten using only supported facts or removed. The Large Language Model is called once for drafting and once for constrained rewriting. The verifier stores claim decisions and evidence identifiers, allowing auditors to inspect why a sentence was changed. This architecture separates generation quality from factual acceptance.

3. Experiments

On FactProbe, the baseline hallucination rate is 18.1% and GraphVerify lowers it to 7.2%. On HotpotQA, Exact Match is 34.8 before verification and 34.6 after verification, showing no meaningful loss. Verification adds 42 ms per answer on average; 91% of removed claims lacked a supporting fact in the Knowledge Graph.

System	FactProbe hallucination	HotpotQA EM
Unverified generator	18.1%	34.8
GraphVerify	7.2%	34.6

4. Conclusion

GraphVerify makes Hallucination Reduction operational by checking every claim after generation. SubGround supplies evidence, the Relational Fact Lattice keeps lookup fast, and the Large Language Model is constrained during repair. The near-constant HotpotQA accuracy and large FactProbe improvement support post-hoc Fact Verification as a practical safety layer.

References

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